

Portfolio Class Cheat Sheet

m Portfolio(name, spec_list, uw=None, label=None, label_map=None, note="", tags=(), doc="")

A Portfolio is a collection of Aggregate units sharing a discretization grid; it adds diversification, capital allocation, and distortion pricing. spec_list is a list of agg specs / programs; usually built via build('port ...'). The following tables show all **m** methods, and fields or properties (used interchangeably). Comments elucidate the meaning of more obscure entries.

1. Specification & creation

name (identity), label (display), agg_list (the units), n_units, unit_names, unit_names_ex (with total), unit_name_pipe, <unit> (each unit by name, e.g. A), value_type, exp_premium, program, pprogram, spec, spec_ex, **m** as_severity (collapse the total to a Severity), **m** create_from_sample, **m** create_frequency

2. Update

log2, bs, sev_calc, discretization_calc, normalize, padding, last_update, hash_rep_at_last_update, **m** update, **m** best_window (choose the shared combine grid), **m** snap, **m** remove_fuzz, **m** trim_density_df, **m** swap_density_df

3. Moments

actual = analytic, est = estimated from the FFT grid
actual_m, actual_cv, actual_sd, actual_var, actual_skew, est_m, est_cv, est_sd, est_var, est_skew, ex ($E[X]$)

4. Statistical functions

m cdf, **m** sf, **m** pdf, **m** pmf, **m** q (VaR), **m** tvar, **m** tvar_threshold, **m** var_dict^[1], prob_eq_0 ($\Pr(X=0)$), **m** sample, sample_df, **m** sample_compare, **m** sample_density_compare

5. Validation

valid, validation_eps, validation_df, validation_explanation, bs_window_df, bs_description, bs_explanation, tail_class (worst of the units), tail_behavior_df, tail_description, tail_explanation, bounded

6. Output dataframes

density_df^[1], density, summary_df (per unit + total, at a glance), stats_df, info, tail_df, **m** tail_periods_df, independent_density_df, independent_stats_df (the no-diversification comparison), **m** unit_density, **m** unit_density_df, **m** aligned_unit_density_df, **m** augmented_df, augmented_dfs (per-distortion cache), figure

7. Reinsurance

reins_kinds, reins_description (look-through, unit by unit), reins_summary_df, reins_density_df, reins_stats_df (*per-unit reinsurance is set on each Aggregate*)

8. Visualization

m plot, **m** scatter, figure

9. Risk and pricing

m price, **m** price_ccoc, **m** price_stand_alone, **m** price_pentagon, **m** price_pentagon_ex, **m** pricing_at, **m** pentagon_at, **m** prob_loss_assets, **m** apply_distortion, distortion, distortions, distortion_df, calibration_df, **m** calibrate_distortions, **m** analyze_distortion, **m** analyze_distortions, allocation_method (linear / lifted), **m** allocation_diagnostics, **m** allocation_bounds, **m** pricing_bounds^[2], **m** bodoff, **m** add_exa, **m** add_exa_sample, **m** make_comonotonic_allocations

10. Approximations

m approximate (fit a single Aggregate to the total)

11. Meta

m help, **m** json, **m** save, **m** format_program, pprogram_html, note, tags, doc (the Decl trailer), label_map, labels, renamer, use_labels

Notes:

[1]: matches Aggregate.

[2]: both return objects from aggregate.bounds; see the Bounds card. allocation_bounds ranges the natural allocation by unit; pricing_bounds ranges the price of any other risk.

Abbreviations: m=mean, cv=coefficient of variation, sd=standard deviation, var=variance, skew(ness); VaR=value-at-risk; actual=analytic, est=FFT estimate; exa = $E[X_i | X]$ (conditional expected allocation); CoC=cost of capital.